

This excerpt from *An Introduction to Statistical Learning* (Section 4.6.3) formalizes the transition from ad-hoc models to the unified framework of **Generalized Linear Models (GLMs)**.

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1. The Core Conceptual Bridge: The Link Function (η)

In all three regression approaches, we predict a response Y conditional on features $X_1, \dots, X_p$ by assuming Y originates from an exponential family distribution:

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- **Linear Regression:** Gaussian (Normal) distribution.
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- **Logistic Regression:** Bernoulli distribution ($Y \in \{0, 1\}$).
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- **Poisson Regression:** Poisson distribution ($Y \in \{0, 1, 2, \dots\}$).
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Instead of modeling the raw response directly as a linear combination $\beta^T X$, a GLM applies a **link function** $\eta(\cdot)$ to the expected value $\mu = E(Y | X)$ such that the *transformed mean* is linear:

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$$\eta(E(Y | X_1, \dots, X_p)) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = \beta^T X$$

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2. Comparison of the Three Canonical Formulations

Regression Family	Distribution of $Y X$	Canonical Link Function $\eta(\mu)$	Mean Representation $E(Y X) = \mu(X)$
Linear	Gaussian	Identity: $\eta(\mu) = \mu$	$\beta^T X$
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Logistic	Bernoulli	Logit (Log-Odds): $\eta(\mu) = \log(1 - \mu)$	$1 + e^{\beta^T X} e^{\beta^T X} = 1 + e^{-\beta^T X} 1$
	PNG	PNG	PNG
Poisson	Poisson	Log: $\eta(\mu) = \log(\mu)$	$e^{\beta^T X} = \exp(\beta^T X)$
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- In **logistic regression**, the logit link maps the probability $\mu \in (0,1)$ to $(-\infty, \infty)$.
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- In **Poisson regression**, the log link maps the positive count rate $\mu \in (0, \infty)$ to $(-\infty, \infty)$, guaranteeing that $\lambda(X) = e^{\beta^T X}$ remains strictly non-negative.
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3. Footnote 5: The Overdispersion Caveat

Footnote 5 flags a crucial real-world consideration:

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"In fact, the variance in the Bikeshare data appears to be much higher than the mean, a situation referred to as overdispersion. This causes the Z-values to be inflated in Table 4.11..."

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- Under a theoretical Poisson model, **equidispersion** is assumed:

$$\text{Var}(Y) = E(Y) = \lambda$$

- In real-world sensor, physical security, or traffic environments, data is almost always **overdispersed** ($\text{Var}(Y) > E(Y)$) due to unmeasured latent factors (bursty weather, shift changes, or coordinated group movements).
- When building production edge anomaly filters, recognizing overdispersion is what separates a naive detector from an industrial one:
 - Overdispersion causes standard Poisson error estimates to be too small, leading to false confidence and inflated Z-scores.
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 - Handling it requires scaling variance via **Quasi-Poisson** estimation or swapping the count kernel to a **Negative Binomial distribution**, which incorporates a dispersion parameter θ to absorb excess variance without breaking runtime compute budgets.

4. Implementation in VectorEdge Edge Silicon

This unification is why VectorEdge can execute low-SWaP statistical classification directly on a tiny ARM Cortex-M or RISC-V edge core:

1. **Uniform Mathematical Form:** Regardless of whether the edge daemon is calculating class probabilities (logistic) or arrival anomaly rates (Poisson), the compute core is evaluating the **exact same linear inner product**:
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$$z = \beta_0 + \sum_{j=1}^p \beta_j X_j$$

2. **Lookup Table Inversion:**

- For classification: $\Pr(Y=1) = \sigma(z)$ (sigmoid).
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- For arrival rates: $\lambda(X) = \exp(z)$.
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- On low-power hardware without a hardware FPU, both $\sigma(z)$ and $\exp(z)$ are calculated via a pre-compiled 1D LUT (lookup table) or fixed-point Taylor expansion in under 100 clock cycles.

3. **No Heavy Matrix Inversion at the Edge:** Model training and iteratively reweighted least squares (IRLS) happen in the cloud via Vertex AI; the edge hardware only evaluates the inner product $\beta^T X$ and inverse link function $\eta^{-1}(z)$.

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